

3.1 OSCILLATIONS P2&P3 QUESTIONS

Compiled by: R. Muga

Q1: NOV 2004 P2

3 (a) Define

(i) simple harmonic motion,

(ii) period of an oscillation,

(iii) angular frequency.

[4]

- (b) Fig. 3.1 shows a folded mosquito net which hangs from a ceiling and rotates in such a way that it describes torsional (twisting) oscillations of period 21 seconds.

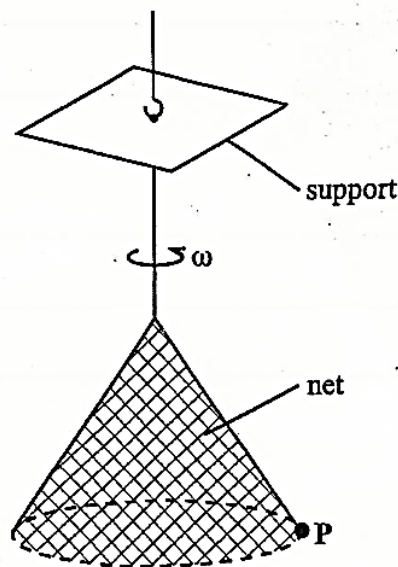


Fig. 3.1

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The angular speed of a point P on the net at any instant is ω .

Q

Calculate

- (i) the frequency of the twisting oscillations,

frequency = _____

[1]

- (ii) the angular frequency of the oscillations.

angular frequency = _____

[1]

- (iii) Explain the relationship between the angular frequency and ω .

[1]

Q2:2006 SPEC P2

- 4 (a) Define *simple harmonic motion*.

[2]

- (b) Fig. 4.1 shows an aluminium blade with a broad end clamped in front of one pole of a magnet.

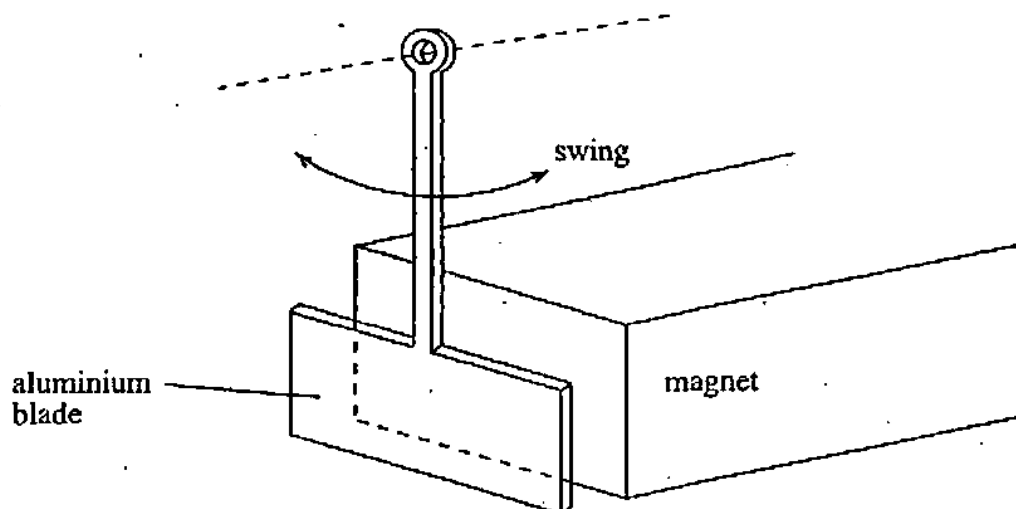


Fig. 4.1

The blade is displaced to one side and released to oscillate in front of the magnet. Damped oscillations are observed.

- (i) 1. Explain why the oscillations are damped.

2. Give a statement of the law that explains this phenomenon.

3. Explain why this law is an example of conservation of energy.

[5]

- (ii) Suggest an improvement that can be made on the aluminium blade to reduce the damping.

- (iii) Explain how this improvement reduces damping.

[3]

Q3: NOV 2007 P2

- 2 (b) In a Science Club, a group of A Level Physics students made a model of the earth and an infinitely long smooth bar resting on it as in Fig. 3.1.

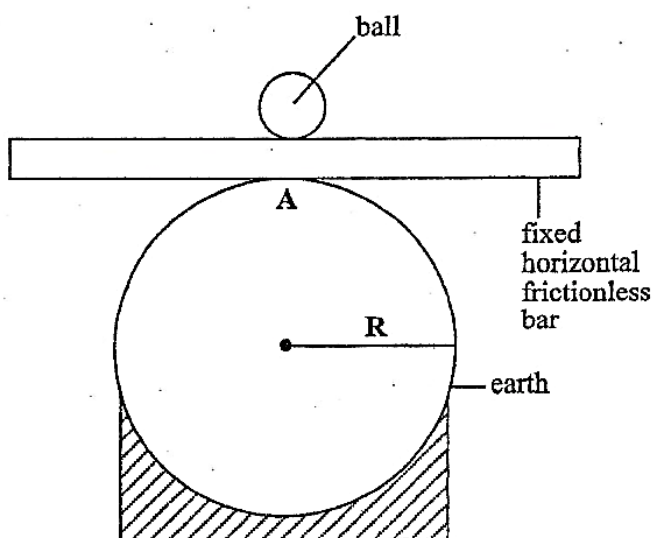


Fig. 3.1

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A smooth ball is placed on the bar as shown in Fig. 3.1. Considering this system to be real, when the ball is given a push, it will either oscillate about A or continue to move to infinity.

- (i) Suggest why it would be possible for the ball to oscillate about A.

- (ii) State one condition necessary for the ball to move to infinity.

[3]

Q4:NOV 2009 P2

- 3 (a) A diver times his bounces on a springboard so as to build up a large amplitude but a comfortable car reduces the amplitude of oscillation in the shortest possible time.

- (i) Identify the phenomenon described for

1. the diver,

2. the car.

[2]

- (ii) Sketch well labelled graphs for each phenomenon.

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Q5:JUN 2010 P2

- 3 (a) State two conditions that must be satisfied by a body executing simple harmonic motion.

1. _____

2. _____ [2]

- (b) The displacement x in cm of the needle of an electric sewing machine is represented by the equation

$$x = 1.5 \cos 2.5\lambda t.$$

Determine

- (i) the amplitude and period of oscillation,

amplitude = _____

period = _____

- (ii) the acceleration when $t = 0.1$ seconds.

acceleration = _____ [5]

Q6: NOV 2011 P2

- 3 Fig. 3.1 shows the variation of velocity with displacement for a 0.200 kg mass.

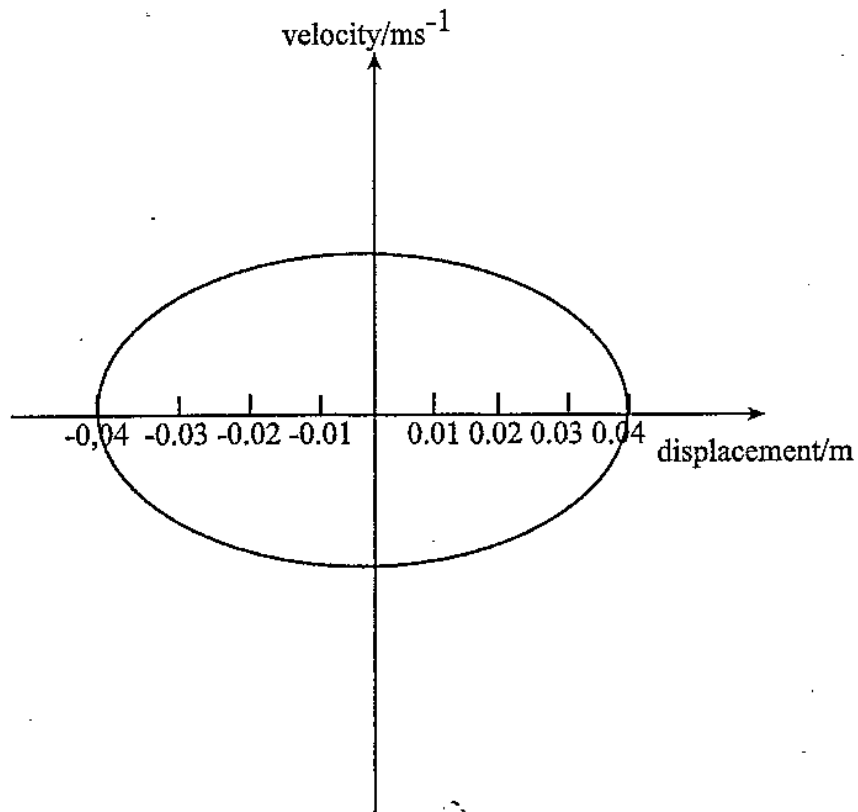


Fig. 3.1

Given that the period of oscillation is 0.050 s, determine

- (i) the frequency, f ,

$$f = \underline{\hspace{2cm}}$$

- (ii) the angular frequency, ω ,

$$\omega = \underline{\hspace{2cm}}$$

- (iii) the maximum velocity, v ,

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- (iv) the maximum kinetic energy.

maximum k.e. = _____

[7]

Q7:JUN 2012 P2

- 3 Fig. 3.1 shows the variation of displacement, x , with time in seconds, t , for a simple pendulum.

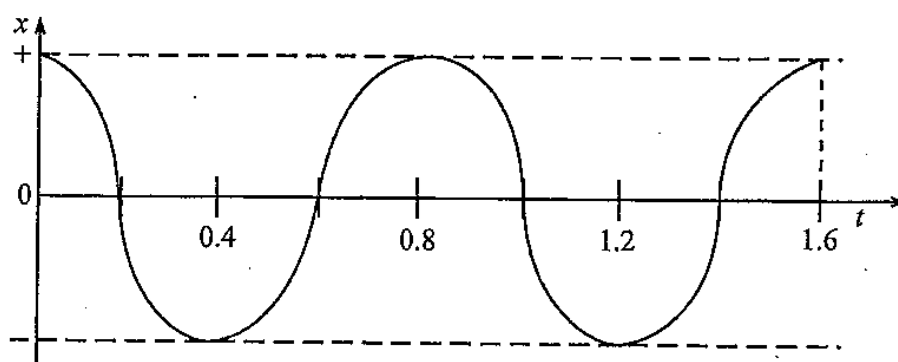


Fig. 3.1

- (a) Determine

- (i) the period,

period = _____

- (ii) the frequency of oscillation.

frequency = _____

[2]

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- (b) Sketch on Fig. 3.1, a graph of the variation of
- (i) velocity with time and label this v ,
 - (ii) acceleration with time and label this a . [4]
- (c) Describe the energy changes that take place from the moment the pendulum is released until it completes half an oscillation.
- _____
- _____
- _____
- _____
- _____
- _____ [3]
- (d) (i) Explain the term *damped oscillations*.
- _____
- _____
- (ii) Give a practical example where damping is useful and state the type of damping.
- example: _____
- _____
- type of damping: _____ [3]

Q8: NOV 2014 P2

- 3 (a) (i) Define *simple harmonic motion*.

- (ii) Fig. 3.1 shows the variation of energy of an oscillating simple pendulum with its displacement.

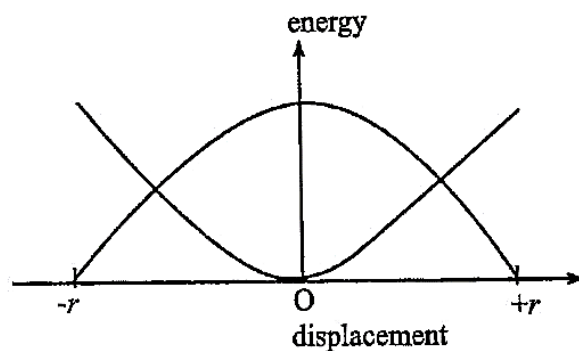


Fig. 3.1

On Fig. 3.1 label the curve for potential energy, E_p , and that for kinetic energy, E_k .

[4]

- (b) (i) Sketch displacement-time graphs to distinguish between light damping, heavy damping and critical damping.

- (ii) Give one practical example where oscillations which are critically damped are observed.

[4]

3.1 OSCILLATIONS P2&P3 QUESTIONS

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Q9:JUN 2016 P2

- 4 (a) State the principle of superposition.

[2]

- (b) Give any three conditions necessary for two-source interference fringes to be observed.

1.

2.

3.

[3]

- (c) In a Young's double-slit experiment, the distance between the slits and the screen is 1.60 m. Using light of wavelength 5.89×10^{-7} m, the distance between the centre of the interference pattern and the fourth bright fringe on either side is found to be 16.0 mm.

Calculate the slit separation.

slit separation =

 [3]

Q10:JUN 2017 P2

- 2 (a) (i) Explain the meaning of the following terms:

1. *free oscillations*

2. *simple harmonic motion*

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- (ii) Explain why in practice the amplitude of a free oscillation is not constant.

[4]

- (b) The position of an object undergoing simple harmonic motion is given by $x = x_0 \sin 0.5t$.

Sketch on the same axis the graphs showing the variation with time of the object's

- (i) speed,
(ii) acceleration.

Q11:NOV 2002 P3

- 2 (a) State two differences between transverse and longitudinal waves. [3]

- (b) Fig. 2.1 shows the graph of a snapshot of the waveform of a water wave. A particle A is 0.60 m from the origin O.

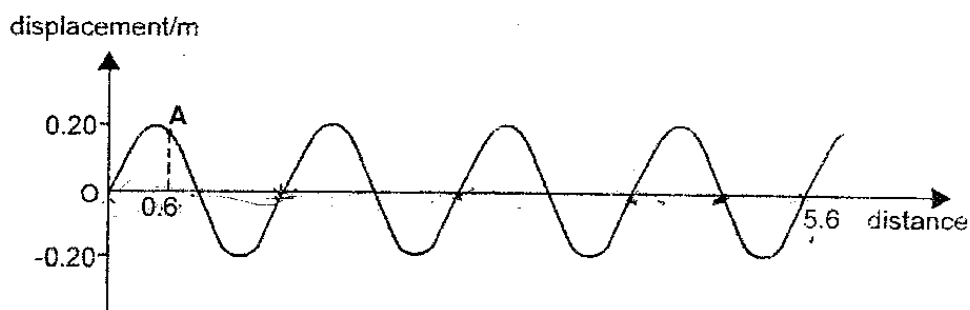


Fig. 2.1

From Fig. 2.1 determine:

- (i) the wavelength,
(ii) the phase difference between particles at O and A.

[2]

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- ✓(c) Fig. 2.2 shows the graphs of displacement against time for the particle A.

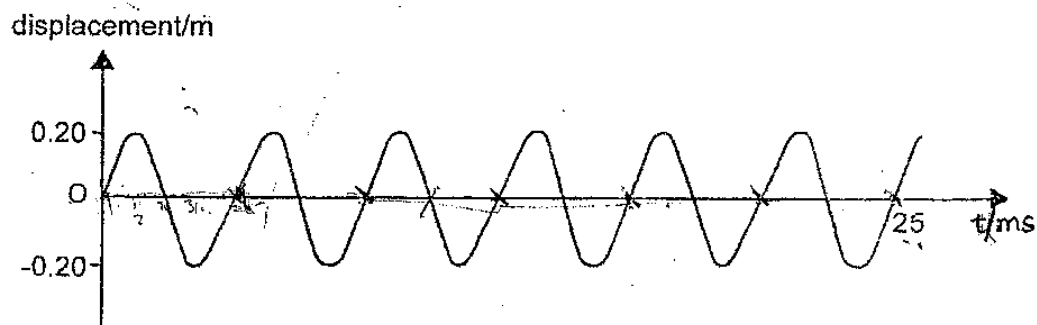


Fig. 2.2

Determine:

- (i) the period, T of the oscillations of particle A,
- (ii) its frequency,
- (iii) the speed of the water waves.

5 - f
[5]

- ✓(d) Given that particle A describes simple harmonic motion, calculate its

- (i) angular frequency,
- (ii) maximum velocity,
- (iii) maximum acceleration.

[6]

- ✓(e) (i) What is meant by *resonance*?

- ✓(ii) Describe **two** situations in which critical damping is employed. [4]

Q12:NOV 2009 P3

- 3 (a) Explain the terms

- (i) angular frequency, and
- (ii) phase difference.

[2]

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- (b) The graphs in Fig. 3.1 are showing oscillations of carbon atoms.

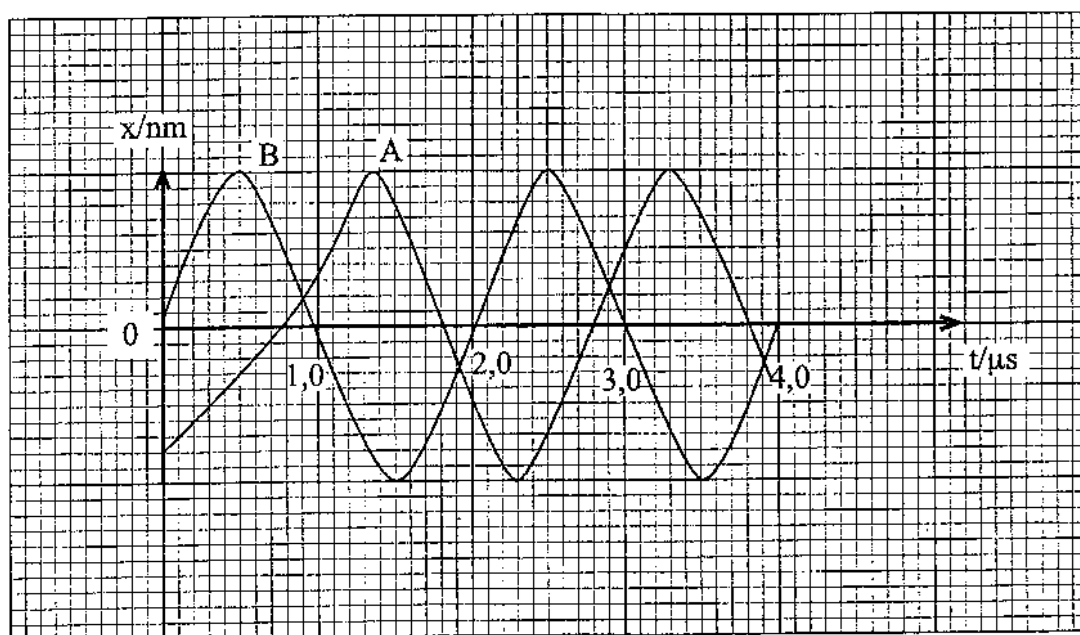
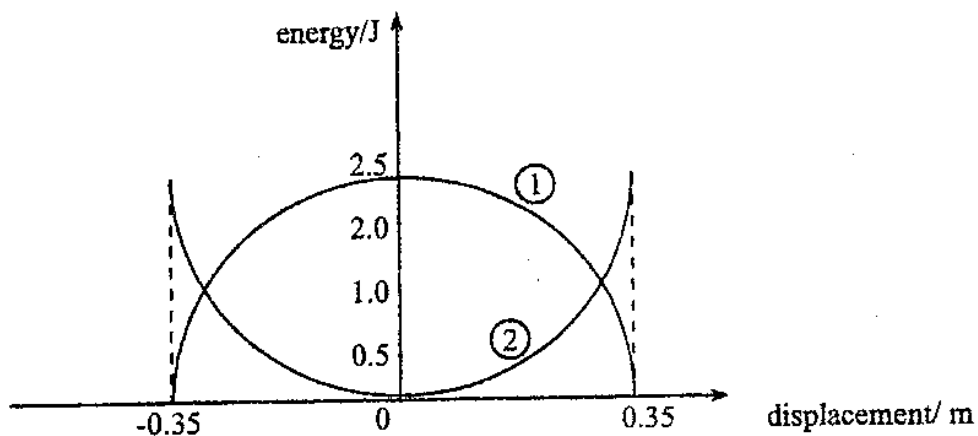


Fig. 3.1

- (i) State with a reason the oscillation that is leading.
 - (ii) Calculate the phase difference in radians between the two oscillations. [5]
- (c) Sketch a graph of velocity against displacement for one of the oscillating atoms. [3]

Q13:JUN 2011 P3

- 1 (c) Fig. 1.2 shows the variation of the *kinetic* and *potential* energies with *displacement* for a particle of mass, 2.0 g, on a spring executing simple harmonic motion.



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- (i) Identify the graphs 1 and 2.
- (ii) Determine the
 1. maximum speed of the particle,
 2. spring constant, k .
- (iii) Explain the effect, if any, of using a driver force of increasing frequency on the oscillations of the particle.

[6]

- 2 (a) (i) Distinguish between *damped* and *forced oscillations*.
- (ii) Describe the effect of damping on resonance.

[4]

Q14:NOV 2012 P3

- 1 (c) Fig. 1.1 shows the variation of potential energy, U , with position, x , of a 4 kg particle executing simple harmonic motion.

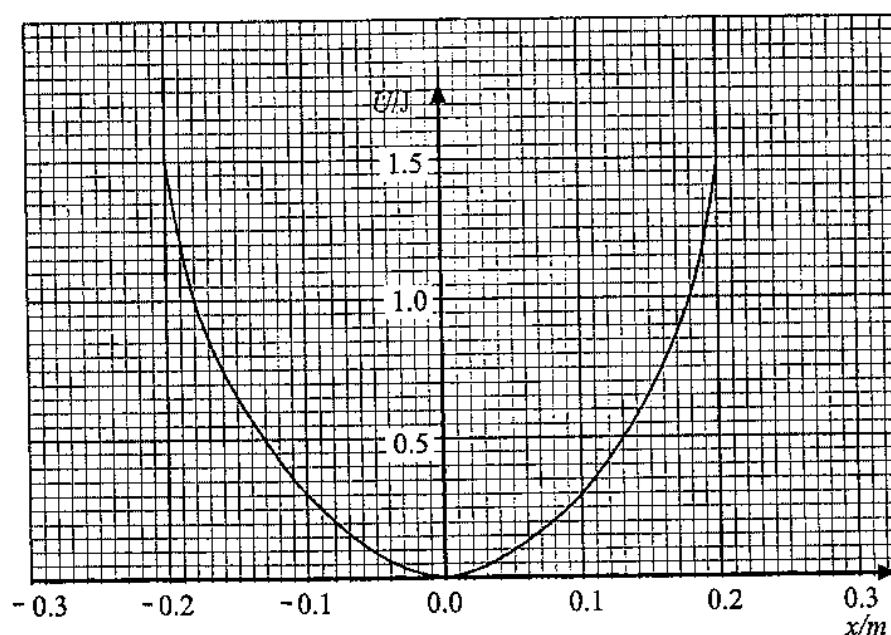


Fig. 1.1

- (i) State the amplitude of oscillation.
- (ii) Determine the period of oscillation.

[4]

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- (d) (i) Explain the term *resonance*.
- (ii) A well with vertical walls resonates with a sound note of frequency 7.0 Hz and does not resonate at lower frequencies.

Estimate the depth of the well, if the speed of sound in air is 350 ms^{-1} .

[4]

Q15:JUN 2013 P3

- 1 (d) (i) Define *simple harmonic motion*.
- (ii) A mass at the end of a helical spring is given a vertical downward displacement of 2.0 cm from its rest position and released.
- If the mass moves with simple harmonic motion of period 2.0 seconds, calculate the displacement of the mass during the first 0.75 seconds.
- (iii) Simple harmonic motion may be used in wall clocks.

Explain why the oscillating system is normally enclosed inside the clock.

[5]

Q16:JUN 2014 P3

- 1 (c) (i) Define *resonance*.
- (ii) Soldiers on a parade march in phase crossing a suspension bridge.
- Given that the soldiers can vary their frequency of marching, sketch a graph to show the variation of the amplitude of the bridge as a result of the frequency of the marching soldiers.

- (iii) Suggest what might occur to the bridge at resonance.

[4]

- (d) A simple pendulum set to oscillate in air eventually comes to rest.

- (i) State the effect of air on the motion of the pendulum.

- (ii) Describe what happens to the kinetic energy of the pendulum as a result of the effect you stated in (i).

- (iii) In everyday life passengers prefer cars with good working shock absorbers in their suspension systems.

Explain why passengers avoid cars with poor shock absorbers.

[4]

Q17:JUN 2015 P3

- 3 (a) State the equation for simple harmonic motion, explaining the significance of the signs and symbols in the equation.

[3]

- (b) A loaded closed test-tube of mass, m , and cross-sectional area, A , floats in equilibrium with length, l , under the surface of a liquid of density, ρ . The test-tube is pressed down in the liquid a distance, x , as in Fig.3.1 and released.

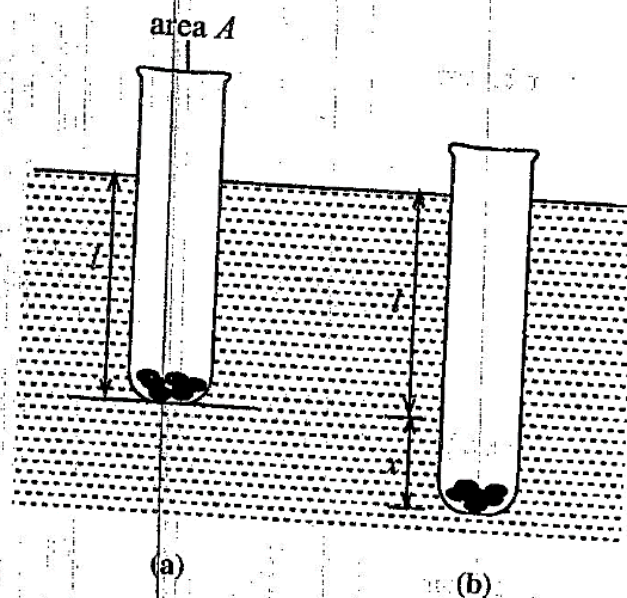


Fig.3.1

- (i) The test-tube oscillates with simple harmonic motion.

Show that acceleration, a , is given by

$$a = \frac{-\rho Ag}{m} x$$

where g is the acceleration of free fall.

- (ii) Determine an expression for the period of the test-tube.

[5]

- (c) In practice the test-tube fails to execute true simple harmonic motion. Explain.

[2]

Q18:JUN 2018 P3

- 3 (b) Fig.3.1 shows how the displacement of an oscillating particle varies with time.

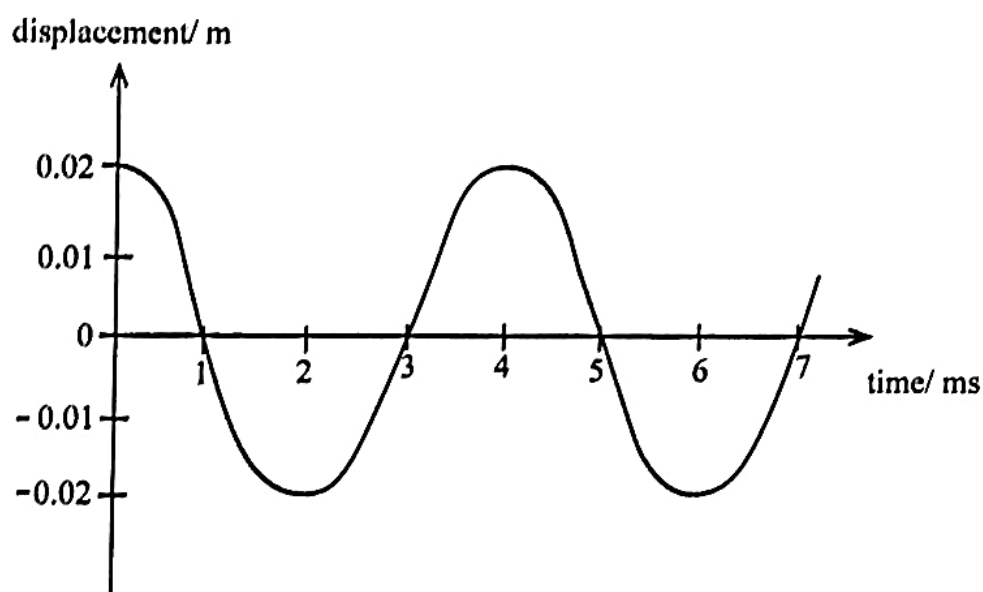


Fig.3.1

- (i) Determine the
1. frequency of the particle,
 2. phase difference between time $t = 0$ ms and time $t = 5$ ms,
 3. maximum acceleration.
- (ii) Sketch a labelled graph to show the variation of acceleration with displacement.

[8]

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Q19: NOV 2018 P3

- 2 (a) (i) Explain the concept of *damping*.
- (ii) A car of mass 1.2 tonnes has four springs and oil dampers in its suspension system. At equilibrium each of the springs is shortened by 50 mm. To test if the springs and dampers are working the car is pressed vertically downwards 30 mm and released. The car executes simple harmonic motion when the dampers are not working and is critically damped when dampers are working.
1. Determine the period of oscillation of the car when dampers are not working.
 2. Sketch two graphs on the same axes to compare the oscillations of the car when dampers are not working and when they are working.
- (iii) Describe energy changes that occur when oil dampers are working.

[7]

Q20: 2018 SPEC P3

- 2 (a) (i) Define *simple harmonic motion*.
- (ii) One end of a spring of spring constant k , is attached to a rigid support. A mass m , hanging from the lower end of the spring is displaced downwards from equilibrium and released.

Show that the mass execute simple harmonic motion.

[5]

Q21: JUN 2019 P3

- 2 (a) (i) Explain what is meant by *resonance*.
- (ii) Describe the effects of
1. forcing frequency on amplitude of oscillation,
 2. the degrees of damping on resonance.

[8]

3.1 OSCILLATIONS P2&P3 QUESTIONS

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Q22:NOV 2007 P3

- (2) (a) Define *simple harmonic motion*. [2]
- (b) Oscillation of a simple pendulum can be considered to be simple harmonic.
State and explain a condition that should be satisfied for the motion to be simple harmonic. [2]
- (c) For the simple pendulum, explain the relationship between the direction of
(i) velocity and acceleration,
(ii) force and the direction of acceleration. [4]
- (d) A simple pendulum of length 0.80 m has an amplitude of 5.0 cm. The mass of the bulb is 40.0 g. Given that the period, T of oscillation is given by,
$$T = 2\pi\sqrt{\frac{l}{g}},$$
where l is the length of the pendulum, calculate
(i) the kinetic energy of the bob,
(ii) its potential energy at a displacement of 3.2 cm. [5]
- (e) Explain why the pendulum finally comes to rest. [3]
- (f) Suggest why in the formation of stationary waves in strings only particular values of frequency will result in loops being formed. [4]